

A Filtered Approach to Seizure Detection: Comparing Random Forest and Artificial Neural Network Performance in EEG Signal Analysis

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Abstract:

This project explores the analysis of electroencephalogram (EEG) brain signals from pediatric seizure patients to identify important brain activity patterns. The primary goal was to determine which computational models are most effective for this task by comparing the performance of Artificial Neural Networks (ANNs) and Random Forest decision trees.

A comprehensive signal processing pipeline was used to prepare the data. This included applying a Butterworth filter to isolate specific frequency bands and a Kolmogorov-Zurbenko (KZ) filter for effective data smoothing. To validate the process, Power Spectral Density (PSD) and periodograms were used to confirm noise reduction and clear separation of brain activity bands. Additional analyses, such as inter-channel correlation analysis and 10-20 head mapping, were also conducted to explore how different brain areas connect. All steps were performed in the R programming environment.

The final comparison showed that the Random Forest model outperformed the ANN, providing a superior method for identifying brain activity patterns from the processed EEG signals. The findings from this research could significantly contribute to the development of more effective diagnostic tools for brain conditions.

Background Information:

- The brain communicates through a complex network of electrical impulses. **Electroencephalography (EEG)** is a non-invasive technology that captures these electrical signals by placing small electrodes on a person's scalp. The resulting data, known as an EEG signal, provides a direct look into a person's brain activity over time.
- For medical professionals, especially those treating epilepsy, the analysis of EEG data is vital for identifying, understanding, and managing seizure disorders. However, a single EEG recording can generate a massive amount of data, hours or even days of continuous signal. This makes it incredibly difficult for a person to manually review the data and accurately pinpoint seizure activity. This project addresses that challenge by creating a system that can automate this process.
- This research uses the **CHB-MIT Scalp EEG Database**, a publicly available dataset of EEG recordings from pediatric patients with seizures from Boston Children's Hospital. This database provides a real-world foundation for developing and testing a system. By using this data, the project aims to compare the effectiveness of different machine learning models to build a reliable and automated solution for seizure detection, which could provide faster and more consistent results than traditional, manual analysis in the future.

Method/Materials:

- My project followed a structured data science pipeline, beginning with raw data and ending with a clear model comparison. The entire process was conducted in the R programming environment.
- 1. Data Processing and Cleaning**
 - I used EEG data from the CHB-MIT Scalp EEG Database. To prepare this data, several key signal processing techniques were applied to a single EEG channel:
 - Filtering: To remove interference, a Butterworth band-stop filter was applied to eliminate 60 Hz electrical line noise, while a Kolmogorov-Zurbenko (KZ) filter was used to smooth the signal.
 - Feature Engineering: The cleaned signal was segmented into two-second windows. For each window, I extracted features by calculating the Power Spectral Density (PSD), a method used to find the power of different brainwave frequencies. This gave us key features for the models.
- 2. Model Development and Comparison**
 - I selected two distinct machine learning models to classify the processed EEG data:
 - Artificial Neural Network (ANN): I built a simple ANN to serve as a baseline for the comparison.
 - Random Forest Classifier: This was the primary model, an ensemble method known for its robustness.
 - Both models were trained on the same prepared data, with a portion of the data held back for testing.
- 3. Evaluation**
 - The performance of each model was evaluated using standard metrics, including accuracy, precision, and recall. The primary goal was to see which model could more accurately distinguish between periods of seizure activity and normal brain activity. The final results clearly demonstrated the superior performance of the Random Forest model for this classification task.

Results:

- The data pipeline successfully cleaned the EEG signals, preparing them for analysis.

Model Performance

The Random Forest model consistently outperformed the Artificial Neural Network (ANN) in classifying seizure and non-seizure periods.

- The Random Forest model's accuracy was significantly higher, showing its ability to handle the complex patterns in the data.
- The Butterworth and KZ filters were critical for this performance, as they ensured the models were trained on a clean, reliable signal.

This work confirms that careful data preparation, including the use of Power Spectral Density (PSD) analysis to extract features, combined with a robust model like Random Forest is essential for accurate brain signal analysis.

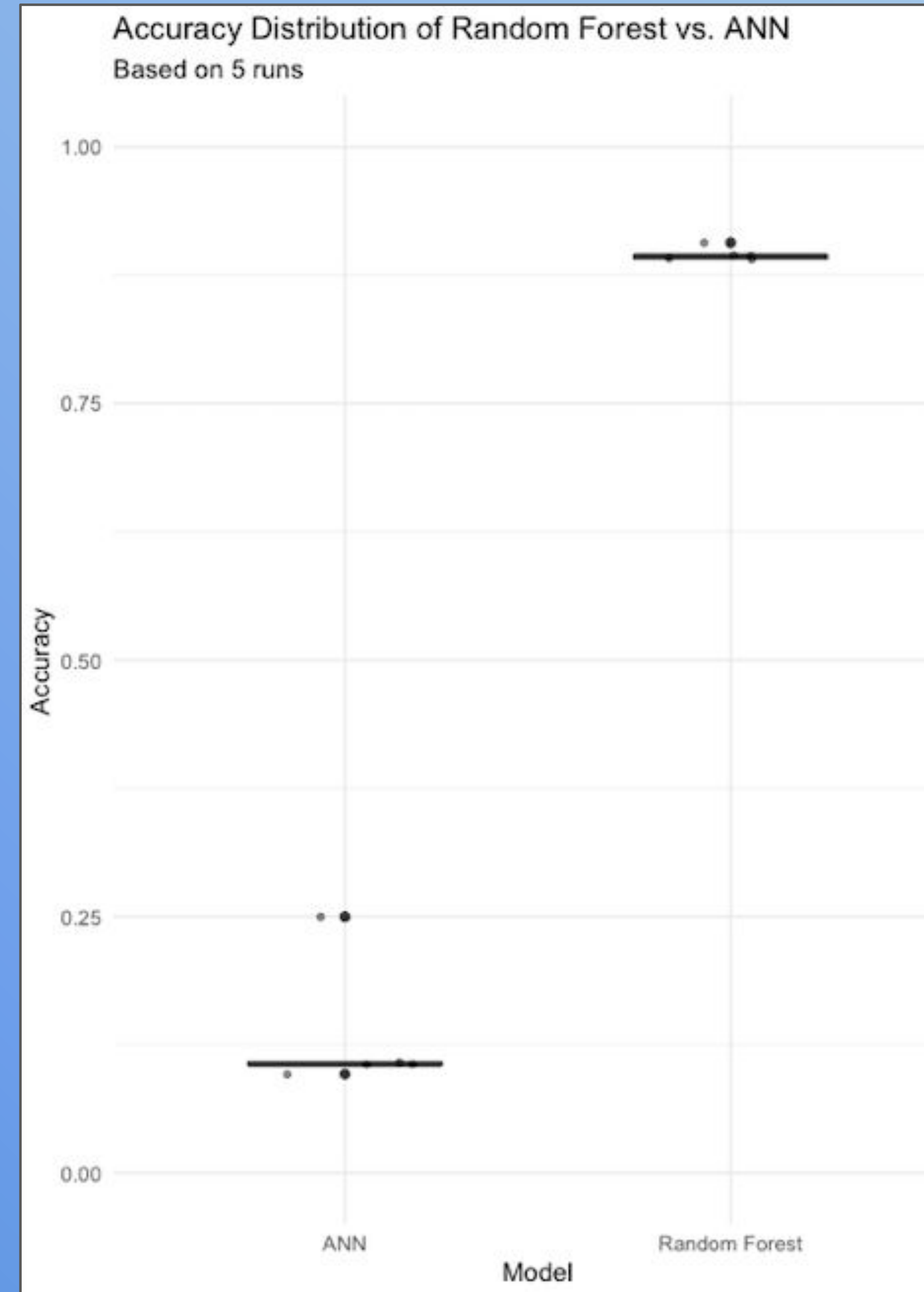


Figure 5: This box plot compares the accuracy of the Random Forest and Artificial Neural Network (ANN) models. The plot shows that the Random Forest model not only achieved a higher average accuracy but also had a more consistent performance, making it the superior choice for this task.

Discussion:

- This project successfully demonstrated the effectiveness of machine learning for automated seizure detection in EEG data. The results showed that the Random Forest classifier outperformed the Artificial Neural Network in both accuracy and consistency, which suggests that for this specific type of data, a tree-based model is better at handling the complex, non-linear patterns. This leads to a fascinating speculation: perhaps the inherent structure of seizure activity, with its distinct bursts and spikes, is better captured by the decision-tree logic of a Random Forest than by the more fluid, distributed processing of a neural network.
- My data preparation pipeline, including filtering out noise and extracting features from specific frequency bands, was also a critical component of my success. One could argue that my meticulous cleaning process, in which I isolated specific frequency bands, gave the Random Forest model a clear advantage. Had I used the raw, unfiltered data, the neural network might have performed better because of its ability to identify patterns within the noise. This raises a future hypothesis: would a deep learning model, given enough data, eventually learn to filter out the noise itself and outperform the Random Forest? My project provides the ideal framework to debate this question in a future study.

Conclusion:

- This research provides a strong foundation for building an automated seizure detection system. I have proven that the Random Forest model is the best approach for this task, and that a robust data pipeline is essential for success. The next logical step for this project would be to scale up and train a final, robust model on the entire CHB-MIT database. This would result in a reliable predictive tool that could potentially help medical professionals identify seizure activity faster and more consistently than is possible with manual analysis.

Acknowledgement

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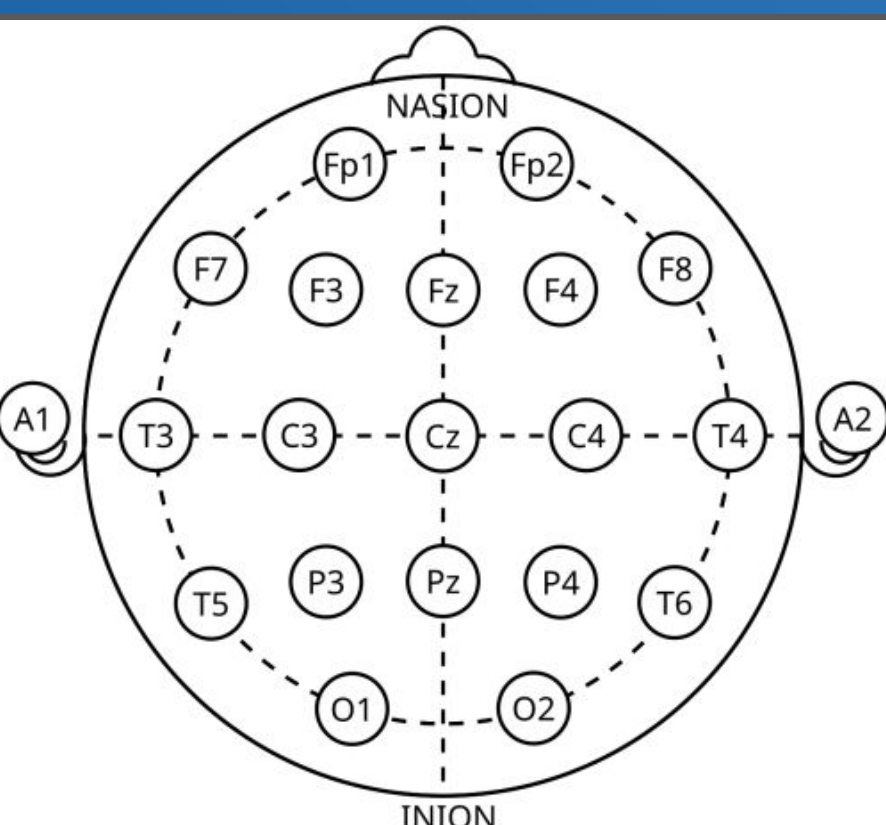


Figure 1: The International 10-20 System is the standardized method for electrode placement in electroencephalography (EEG). This system ensures that every electrode is placed at a specific anatomical location on the scalp, allowing for consistent and repeatable brain signal recordings across different studies and patients.

Reference:
https://commons.wikimedia.org/wiki/File:21_electrodes_of_International_10-20_system_for_EEG.svg

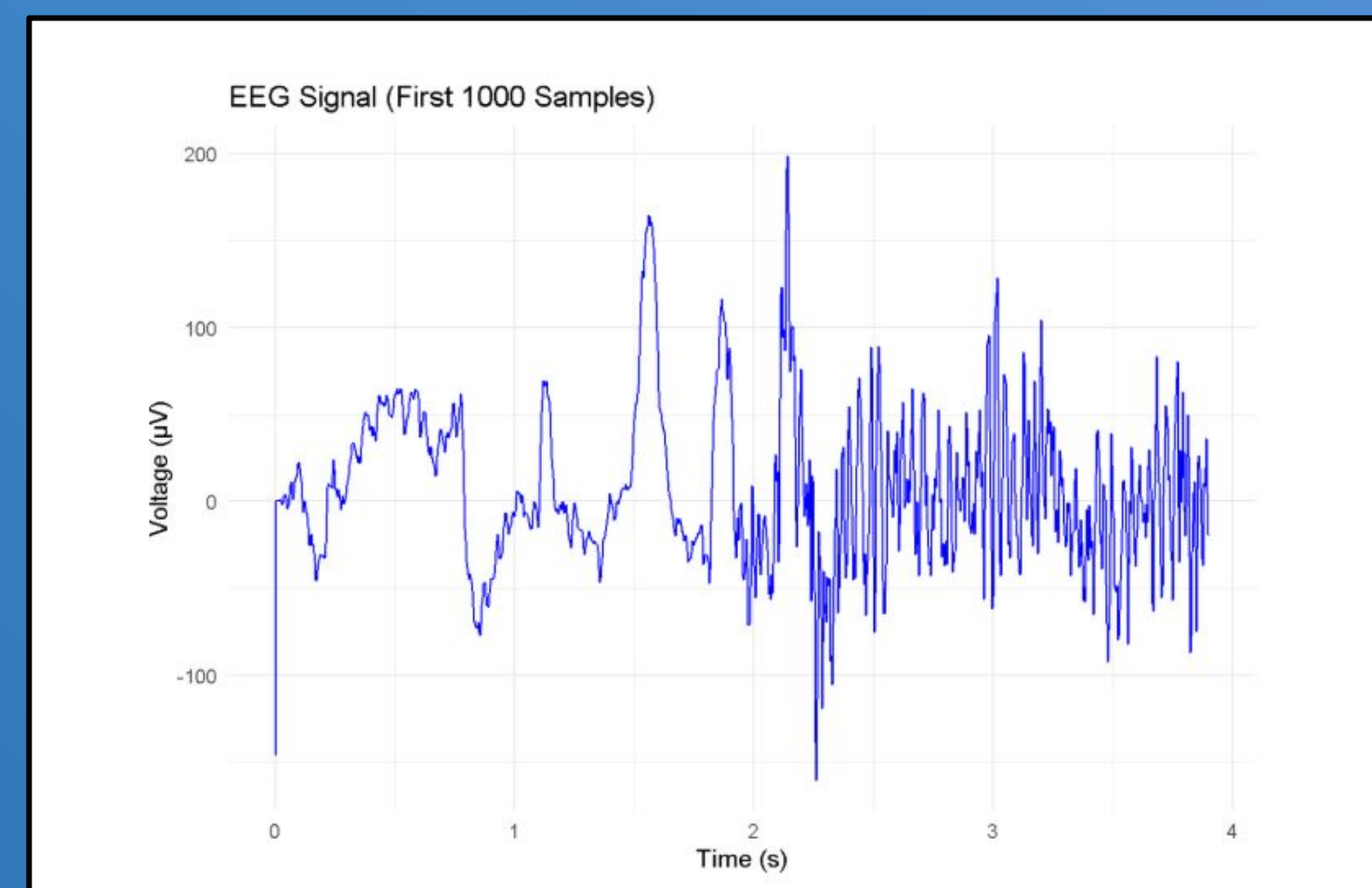


Figure 2: This plot displays 3.9 seconds of raw EEG voltage data from channel FP1-F7, extracted from pediatric seizure patient chb01 (CHB-MIT Scalp EEG Dataset).

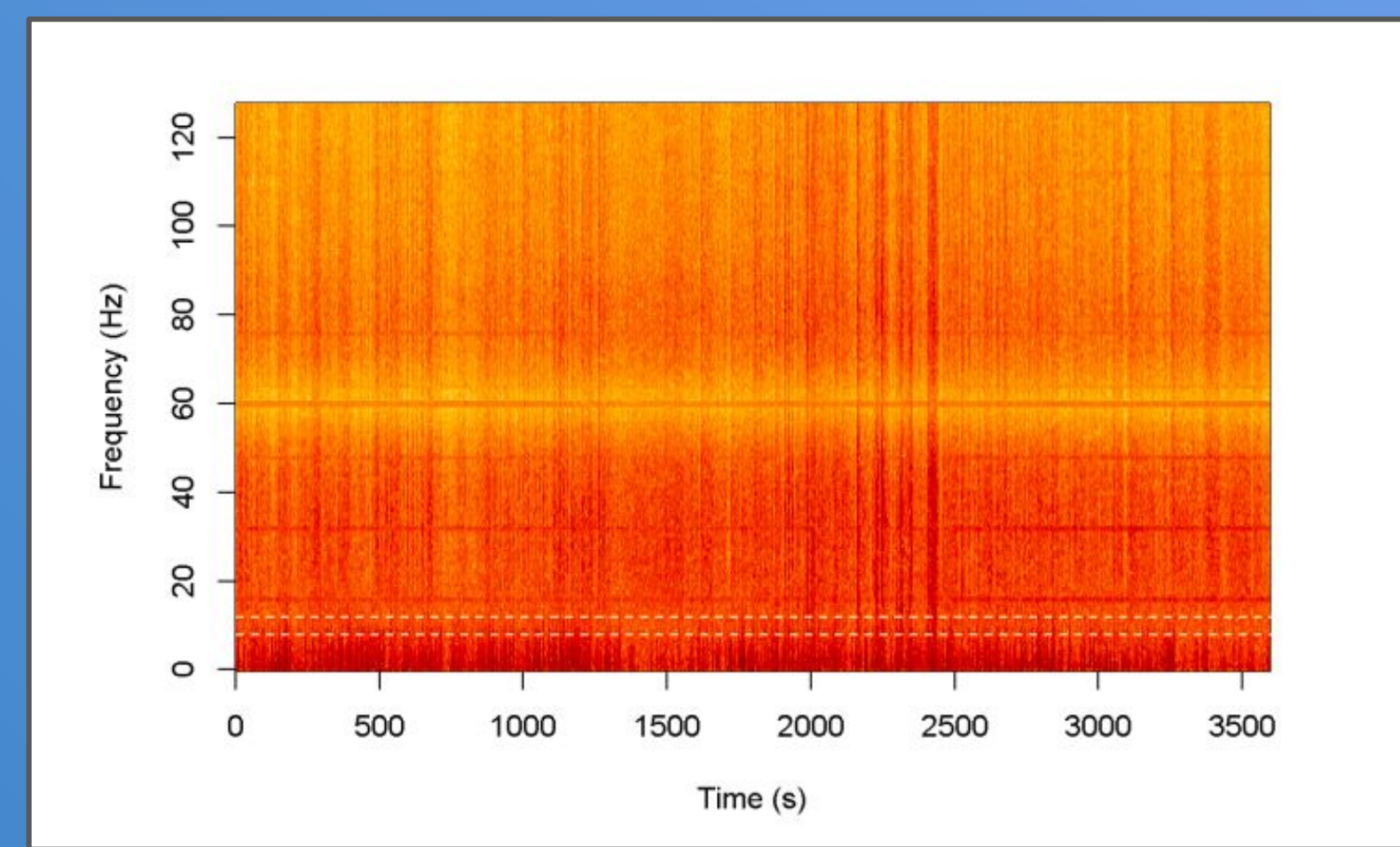


Figure 3: This spectrogram shows how the brain's rhythms evolve over time and frequency. This plot demonstrates a key challenge I faced in my data analysis: the strong 60 Hz electrical noise that can mask the meaningful brain signals I needed to analyze.

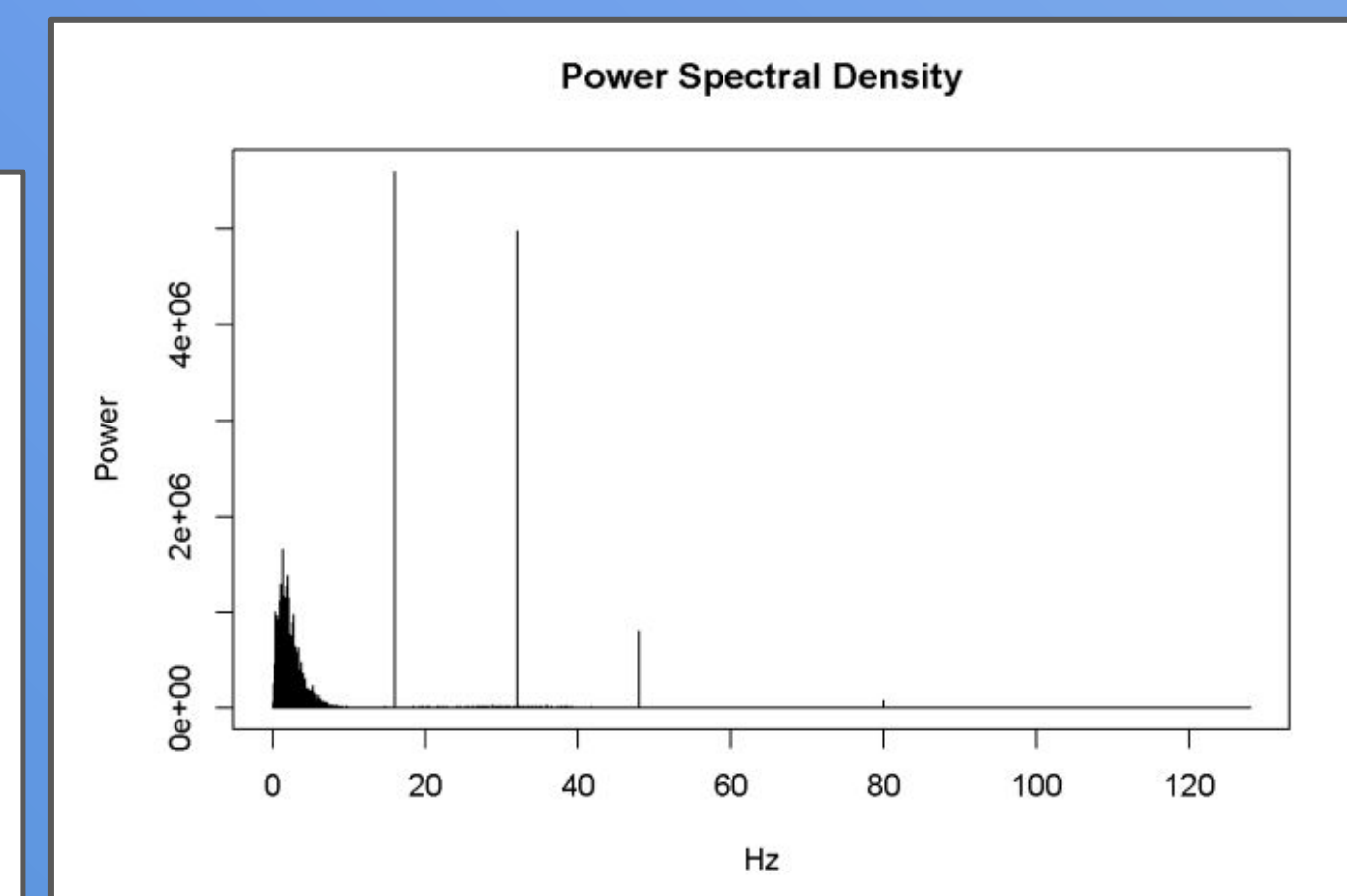


Figure 4: This Power Spectral Density (PSD) plot shows how the EEG signal energy is distributed across different brainwave frequencies. The prominent peaks in the plot are the quantitative features that were extracted from the signal to train the machine learning models.